

Joining Vessel Identity Tables

Below are the methodologies to join the core identity table to each of the feature tables to get information for a specific identity. Beware of issues with NULL values when joining tables. Depending on what language/library/technology used to do the join, you may need to replace NULLs with a placeholder string such as “null” or “no_value” to ensure a proper join.

Authorization table

- Join using all of the following fields: vessel_record_id, ssvid, n_shipname, n_callsign, imo, and flag.

Ownership table

- Join using all of the following fields: vessel_record_id, ssvid, n_shipname, n_callsign, imo, and flag.

Data Caveats

Limit on public identity data

We recognize that the quantity and quality of our collected registry data vary by flag State, thereby introducing inherent bias in information about vessel identity. For instance, we have less complete information about Asian fleets as a smaller number of these flag States make their vessel registries publicly available compared to European nations. While the lack of information is partly complemented by the other types of data sources (self-reported AIS data, machine learning model inference, and vessel monitoring system data published by national authorities), far broader registry data are inevitably required to address the disparity in identity information. Fortunately, a growing number of flag States make their vessel identity data publicly available through global initiatives such as [FAO's Global Record of Fishing Vessels, Refrigerated Transport Vessels and Supply Vessels](#). In addition, we continue to search and incorporate more data sources that are scattered and fragmented in the public domain to provide the public with easy access to information.

Bias on vessel size

Our data has a bias toward larger vessels, in particular, over 24 meters as they are more likely registered to national or international registries than smaller vessels. Apart from a few national-level registries that include vessels as small as 15 meters (e.g. European fleet register and U.S. vessel registry), most registries collected provide limited identity data for small vessels. Additionally, these small vessels are unlikely to be equipped with AIS devices due to lack of relevant regulation or government support, which limits our ability to produce AIS-registry matched records. An increased number of initiatives, including Taiwan Fisheries Agency's financial support in AIS installation, encourage a broader use of AIS or similar tracking

systems, and assist nations in publicly sharing their proprietary vessel monitoring system data to bring greater transparency to small-scale fleets.

Limit on AIS for vessel tracking

Tracking vessel activity through AIS data has several limitations well documented in Kroodsmas *et al.* (2018) and Taconet *et al.* (2019). These limitations hinder our ability to tackle IUU fishing as the vessels that engage in IUU fishing are more likely to keep their transponders off. In some cases, these “dark vessels” have perpetrated large-scale illegal fishing (Park *et al.* 2020).

Emerging technologies and new approaches to address such gaps have been introduced to tackle IUU fishing. While some of them provide useful information about fleet-level characteristics, they are often unable to offer vessel-level characteristics such as vessel identity, owner, and authorization. These vessels are also unlikely to reveal their identity information in public registries, making it even more difficult to investigate the cases. Recently, flag States have adopted more proactive policies that help tackle IUU fishing and punish illegal operators.

Limit on universal standard

Our vessel classification is limited as it does not consider all possible types of vessels around the world. Also, some vessel classes are interpreted differently across states and regions. To ensure the broadest use of our data, however, we are working toward aligning our vessel types, fishing gears, and nomenclature more closely with those adopted by the UN Food and Agriculture Organization. Unique vessel identifiers are also still lacking. IMO numbers are assigned to a limited number of vessels due to lack of relevant policies encouraging a broader group of fishing vessels to adopt an IMO number. In this study, we developed a Vessel Record ID scheme to fill in such a gap. It is, however, dependent on the availability of public registries, and requires more effort to maintain. This ID scheme is temporary and may be subject to change over time. We are currently developing a more consistent ID scheme and upon completion will release an updated version of the data. The number of vessels having adopted IMO numbers continues to grow in recent years, and more flag States and international bodies encourage vessels to apply for an IMO number for greater transparency. This trend will facilitate more robust tracking of the identity history of vessels.